T7 - Multi-system General Trauma Care: Adult and Pediatric
Revised: December 15, 2018

This protocol is to be implemented whenever a trauma patient is identified with obvious or possible multisystem trauma. Patients with injuries capable of causing shock require aggressive early care. Any patient who is injured and has a resting tachycardia is in early shock until proven otherwise.

Emergency Rescue and Rapid Extrication in accordance with current International Trauma Life Support recommendations may be utilized as needed. These runs require medical review when utilized.

LOAD AND GO SITUATIONS.

a. Airway obstruction that cannot be quickly relieved.
b. Hemorrhage that cannot be controlled

COMMUNICATE EARLY. Contact the hospital as soon as feasible of a Trauma Alert. It is important for the hospital to mobilize its trauma service to care for the patient. Keep information brief so the hospital can prepare and you can care for the patient. A full report can be given upon arrival. Include only the following pertinent information:

Mechanism of injury
Injuries
Vital signs and assessment findings
Interventions
Time to arrival

RAPID TRANSPORT. It must be remembered that trauma patients will frequently require surgery as the only chance for survival. Patients must be in the field the shortest possible time to increase survival. Patients who are in extremis should be put in the ambulance and treated enroute. All scenes are different, but do not delay transportation for surgical trauma candidates or load and go patients. The goal, exclusive of extrication time, is to keep scene times down to 10 minutes or less for the multisystem trauma patient. It is often possible to be on the scene for five minutes and accomplish this protocol.

In any pregnant female ≧ 20 weeks gestation or the fundus can be palpated at or above the umbilicus, ensure the uterus is displaced to the left.

Refer to Spinal Motion Restriction Protocol for extrication and management.

Perform the Primary, On Going and Secondary Trauma Assessments in accordance with current ITLS recommendations. Stop hemorrhage immediately. Treat all life threats as you find them. Initiate oxygen, airway maneuvers and adjuncts as needed for respiratory difficulty. Apply necessary components of Basic Initial Medical Care.

Helmets may present management problems for rescuers. Helmets shall be managed in accordance with current trauma standards of care. Exceptions to these standards may be necessitated based on individual needs.

Maintain the patient's body temperature during extrication, treatment and transport.

Bleeding Control:

- Attempt to control **all massive or life-threatening bleeding** with direct pressure initially unless operating in rescue mode within a warm zone (refer to Warm Zone Operations protocol
- If bleeding cannot be controlled by direct pressure to an extremity, apply the **commercial tourniquet** at the highest point on the extremity and adjust until bleeding is stopped. The tourniquet is especially beneficial if the skin edges cannot be approximated. A second tourniquet below the first one may be applied if bleeding continues. Note time of tourniquet application in EPCR and on tourniquet if possible. Do not allow anything to conceal the tourniquet.
- For uncontrollable bleeding on the extremity, head or torso, **Hemostatic gauze or the IT Clamp** (alone or in conjunction with each other) may be applied. The gauze needs to be applied directly to the bleeding site and the wound packed completely, followed by direct pressure for three minutes and then a pressure dressing.
- When utilizing the IT clamp, place additional devices for larger wounds (wounds longer than you thumb~~)~~ Remove or reposition the IT Clamp if~~-~~the device was inadvertently closed prior to placemen; the device was positioned incorrectly; or the patient is ready for definitive surgical wound repair. **THE IT CLAMP IS NOT TO BE APPLIED WHERE DELICATE STRUCTURES ARE WITHIN 10MM OF THE SKIN SURFACE, SUCH AS THE ORBITS OF THE EYES.**

Airway/breathing problems:

1. Tension Pneumothorax
   a) The patient may exhibit the following; anxiety, increasing dyspnea, tachycardia, hypotension, diminished or absent breath sounds on the affected side, unequal expansion of the chest, subcutaneous emphysema, distended neck veins unless patient is hypovolemic, tracheal deviation (a late sign).
   b) Perform a needle chest decompression through the 4th intercostal space, anterior axillary line, over the 5th rib of the affected side. If some contraindication prevents your use of this site, insert needle in the 2nd ICS, mid clavicular line over the 3rd rib on the affected side. Tape the catheter in place.
      Adult Utilize a 12-14 gauge by 3.25 inch catheter
      Child ≤ 20 kg use 18 gauge 2.5 inch decompression needle
      Child > 20 get 14 gauge 3.25 inch decompression needle
2. Open chest wound
   Cover any open chest wounds with 3-sided occlusive dressing or vented chest seal.
3. Flail segment
   Stabilize a flail segment with a bulky dressing.

Abdominal Problems:

Evisceration

Cover with moist saline dressing.

Musculoskeletal Problems:

1. Isolated femur fracture
   a) Apply traction splint if closed fracture.
   b) Utilize splint without traction if fracture is open.
2. Bilateral femur Fractures
   Utilize scoop for movement. Remove patient from scoop once on stretcher.
   May use sager and apply traction if closed fractures to stabilize both legs.
   May use sager without traction if open fractures.
3. Suspected Pelvic fracture(s)
   Apply sheet or pelvic binder to stabilize the pelvis prior to movement.
   Utilize scoop for movement. Remove from scoop once patient is on the stretcher.

Impaled Objects:

Stabilize with bulky dressings to secure the object. Only remove if they interfere with airway maintenance or CPR.

Stabilize an impaled object and place a cup or other protective eye device if possible. Do not apply pressure. Cover unaffected eye with gauze.

Shock:

See Hypotension/Shock protocol